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/* =====
SQL Beginner Practice Project
Student Management System
Topics Covered:
Database Creation
Table Creation
Constraints
Insert
Select
Where
Update
Delete
Default
Timestamp
Sorting
Limit
===== */

/* Question 1:
Create a database called school_db
*/

CREATE DATABASE school_db;

/* Question 2:
Switch to school_db database
*/

USE school_db;

/* Question 3:
Show all tables present in database
*/

SHOW TABLES;

/* Question 4:
Create a students table with:

id -> primary key

name -> not null

```

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email -> unique

age -> must be >=18

status -> default active

created_at -> auto timestamp

updated_at -> auto update timestamp
*/

CREATE TABLE students(

id INT PRIMARY KEY,

name VARCHAR(50) NOT NULL,

email VARCHAR(100) UNIQUE,

age INT CHECK(age>=18),

status VARCHAR(20) DEFAULT 'active',

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

updated_at TIMESTAMP
DEFAULT CURRENT_TIMESTAMP
ON UPDATE CURRENT_TIMESTAMP

);

/* Question 5:
Insert 3 students
*/

INSERT INTO students(
id,
name,
email,
age
)

VALUES

(1, 'Ram', 'ram@gmail.com', 20),

```

```
(2,'Sam','sam@gmail.com',22),  
(3,'Ravi','ravi@gmail.com',19);
```

```
/* Question 6:  
Display all students  
*/
```

```
SELECT *  
  
FROM students;
```

```
/* Question 7:  
Display only name and email columns  
*/
```

```
SELECT  
name,  
email  
  
FROM students;
```

```
/* Question 8:  
Display students whose age > 20  
*/
```

```
SELECT *  
  
FROM students  
  
WHERE age>20;
```

```
/* Question 9:  
Update age of student whose id=1  
Change age to 25  
*/
```

```
UPDATE students  
  
SET age=25
```

```
WHERE id=1;
```

```
/* Question 10:  
Delete student whose id=3  
*/
```

```
DELETE FROM students
```

```
WHERE id=3;
```

```
/* Question 11:  
Try inserting NULL name  
Observe NOT NULL constraint error  
*/
```

```
-- Should Fail
```

```
INSERT INTO students(  
id,  
name,  
email,  
age  
)
```

```
VALUES(
```

```
4,
```

```
NULL,
```

```
'test@gmail.com',
```

```
20
```

```
);
```

```
/* Question 12:  
Try inserting duplicate email  
Observe UNIQUE constraint error  
*/
```

```
-- Should Fail

INSERT INTO students(

id,

name,

email,

age

)

VALUES(

4,

'Kumar',

'ram@gmail.com',

20

);


/* Question 13:
Insert student without status column
Observe DEFAULT value
*/

INSERT INTO students(

id,

name,

email,

age

)

VALUES(

3,
```

```
'Kumar',  
  
'kumar@gmail.com',  
  
23  
  
);  
  
/* Question 14:  
Update name and observe updated_at changes automatically  
*/  
  
UPDATE students  
  
SET name='Naveen Kumar'  
  
WHERE id=3;  
  
/* Question 15:  
Sort students by age descending  
*/  
  
SELECT *  
  
FROM students  
  
ORDER BY age DESC;  
  
/* Question 16:  
Display only first 2 rows  
*/  
  
SELECT *  
  
FROM students  
  
LIMIT 2;  
  
/* Bonus Challenge:
```

Display students

age >18

sort descending

show only first 2 rows

*/

```
SELECT *
```

```
FROM students
```

```
WHERE age>18
```

```
ORDER BY age DESC
```

```
LIMIT 2;
```